

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q22-1989

Wahtoke 138 kV
January 22, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-1989. The Interconnection Customer proposes a 106.3 MW battery energy storage facility interconnecting at the Wahtoke 138 kV substation in the Pennsylvania area. The request was received on January 22, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 689 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$868,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: MIDDLE FORK to HULBERT 138 kV L
Contingency: WASHBURN to FILER 138 kV
Loading: 113.2% (Rating: 283 MVA)
DFAX: 71.2%
Pre-project loading: 86.4%
2. Facility: SOUTHSIDE GENERATING STATION to HULBERT 138 kV
Contingency: FORBING RD to WHILLOCK SUBSTATION 138 kV line
Loading: 103.7% (Rating: 314 MVA)
DFAX: 20.9%
Pre-project loading: 96.6%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
MIDDLE FORK SUBSTATION to HULBERT 138 kV	Rebuild line with higher-capacity conductor	\$846,000	44% (\$372,000)
SOUTHSIDE GENERATING STATION to HULBERT 138 kV	Reconductor line	\$2,874,000	17% (\$496,000)
	Total allocated to project		\$868,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.